

HEINO, Netherlands

WMO: 062780

Lat: 52.434N

Lon: 6.259E

Elev: 16

StdP: 14.69

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	19.1	22.6	13.6	11.0	21.5	17.9	13.6	23.9	25.3	49.5	22.7	48.9	5.3	70	0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.0	83.6	67.9	79.6	66.5	76.1	64.8	70.0	80.1	68.1	77.1	66.4	73.8	6.8	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	97.8	74.0	64.8	92.0	71.7	63.4	87.3	70.1	34.0	80.6	32.4	77.1	31.0	74.1	79.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 19.8	16.9	14.7	DB	14.1	91.8	6.1	3.7	9.7	94.4	6.1	96.6	2.6	98.7	-1.8	101.4	(4)
(5)			WB	13.5	74.8	6.1	2.1	9.1	76.3	5.5	77.5	2.1	78.7	-2.3	80.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.3	37.2	38.5	42.6	48.9	55.4	60.5	63.9	63.1	58.2	51.5	44.3	38.7	(6)	
	DBStd	11.39	7.58	7.05	6.41	6.37	6.37	5.44	5.33	4.69	5.01	6.16	6.07	7.85	(7)	
	HDD50	1679	399	325	239	97	19	0	0	0	2	57	188	353	(8)	
	HDD65	5492	862	743	694	484	304	157	87	93	212	420	621	815	(9)	
	CDD50	1786	1	2	10	63	188	314	430	407	248	103	16	4	(10)	
	CDD65	124	0	0	0	0	7	21	52	35	8	1	0	0	(11)	
	CDH74	1234	0	0	0	16	110	241	495	301	69	2	0	0	(12)	
	CDH80	315	0	0	0	1	16	60	147	80	11	0	0	0	(13)	
(14) Wind	WSAvg	6.9	8.5	8.2	7.8	6.8	6.7	6.1	6.0	5.5	5.6	6.7	7.0	8.2	(14)	
(15) Precipitation	PrecAvg	33.3	2.9	2.4	2.2	1.8	2.4	2.8	3.6	3.4	2.9	2.9	3.0	3.0	(15)	
	PrecMax	44.7	5.2	4.8	4.6	3.7	4.9	6.1	8.2	8.5	6.5	8.3	5.8	6.7	(16)	
	PrecMin	24.8	0.4	0.7	0.4	0.0	1.1	0.5	1.0	0.6	0.3	1.0	0.4	0.9	(17)	
	PrecStd	4.9	1.3	1.2	1.3	0.9	1.0	1.3	1.5	1.9	1.5	1.4	1.2	1.4	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.2	57.7	64.7	76.6	82.0	86.9	90.4	87.9	81.0	71.2	61.0	56.2	(19)	
		MCWB	52.0	49.8	53.9	60.1	65.0	69.5	69.9	69.8	68.0	62.9	56.8	53.0	(20)	
	2%	DB	52.1	53.1	59.6	70.0	76.8	80.5	84.0	81.4	75.4	65.9	57.7	54.0	(21)	
		MCWB	49.6	49.0	51.4	57.5	61.5	66.5	68.4	67.9	65.8	60.3	54.9	51.6	(22)	
	5%	DB	49.6	50.8	55.4	65.4	72.8	75.9	79.4	76.8	71.1	62.6	55.5	51.7	(23)	
		MCWB	47.5	47.5	49.9	55.1	60.3	64.8	67.0	66.0	63.3	58.1	52.5	49.4	(24)	
	10%	DB	47.3	48.3	52.5	61.1	68.3	71.9	75.5	72.9	67.4	60.1	53.1	49.1	(25)	
		MCWB	45.3	45.4	48.1	53.2	58.4	62.6	65.3	64.2	61.0	56.6	50.7	47.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.4	52.0	55.5	61.7	67.4	71.7	72.8	72.2	69.1	63.6	57.7	54.0	(27)	
		MCDB	54.2	54.7	61.0	73.1	78.5	82.3	83.5	84.2	78.2	69.6	59.7	55.3	(28)	
	2%	WB	50.1	49.8	52.8	58.6	64.2	68.3	70.1	69.3	66.5	60.9	55.4	51.9	(29)	
		MCDB	51.8	52.2	57.7	68.3	73.3	77.4	80.6	78.2	73.9	65.1	57.7	53.9	(30)	
	5%	WB	47.6	47.7	50.8	56.1	61.6	66.0	68.3	67.2	64.2	59.0	53.1	49.7	(31)	
		MCDB	49.3	50.4	55.0	63.6	70.2	73.7	77.7	75.0	70.0	62.0	55.2	51.5	(32)	
	10%	WB	45.5	45.6	48.5	53.9	59.3	63.7	66.3	65.3	62.1	57.2	50.9	47.3	(33)	
		MCDB	47.2	48.2	52.0	60.0	66.8	70.3	73.5	71.4	66.4	59.7	52.9	49.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	10.5	13.8	18.2	18.1	17.7	17.0	17.0	16.2	13.0	9.6	8.2	(35)	
		MCDBR	8.9	13.2	20.0	25.9	25.6	25.0	25.1	24.4	21.9	15.2	10.4	8.4	(36)	
	5% WB	MCWBR	8.1	9.9	13.2	14.7	13.0	13.0	11.9	12.7	12.9	10.0	8.4	7.6	(37)	
		MCWBR	9.1	11.7	17.9	23.1	22.3	21.6	22.2	21.5	19.5	14.4	10.1	8.7	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.329	0.351	0.380	0.408	0.411	0.408	0.420	0.412	0.389	0.364	0.343	0.324	(40)	
		taud	2.331	2.325	2.255	2.223	2.253	2.275	2.252	2.294	2.350	2.389	2.354	2.320	(41)	
	Ebn at Noon	Ebn at Noon	207	238	254	261	267	268	262	257	249	230	199	186	(42)	
		Edh at Noon	20	26	34	40	41	40	41	37	31	25	19	17	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	214	429	795	1289	1573	1667	1596	1336	968	560	255	166	(44)	
	RadStd	RadStd	24	75	102	134	129	121	192	110	104	53	26	18	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (74 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page